Appendix

1. Algorithm workflow

Algorithm 1 Training Algorithm

Input: Incomplete multi-view dataset $\left\{\mathbf{X}^v\right\}_{v=1}^V$; Number of cluster K; Maximum epochs E1 and E2; Trade-off parameters λ_1, λ_2, λ_3.
Output: Predicted clustering results **Y**.
Stage I: Representation Pretraining
for e =1 to E1 **do**
Pretraining $\left\{E^v(\cdot),\ D^v(\cdot)\right\}_{v=1}^V$ with Eq. (2)
end for
Output: The shared embedding **Z**.
Stage II: EM-guided Joint Optimization
for e = 1 to E2 **do**
E-step: Perform K-means clustering on shared representation **Z** to update pseudo-labels $p(k|\mathbf{z}_i)$;
M-step: Optimize the entire network by minimizing the objective in Eq. (20)
end for
Final Inference: Obtain the final clustering labels **Y** by applying K-means to the completed latent embeddings.

2. Implementation Details

To ensure reproducibility, we provide the complete network architecture and implementation details of the proposed framework. The autoencoder consists of a four-layer encoder with dimensions Input → 500 → 500 → 300 → 64 and a symmetric decoder with dimensions 64 → 300 → 500 → 500 → Input. The learned latent representation is decomposed into two disentangled components: a 32-dimensional shared semantic factor that captures view-invariant information and a 32-dimensional view-specific factor that preserves view-dependent characteristics. To model the latent distribution, we employ a conditional normalizing flow composed of 10 affine coupling layers. Each coupling layer is parameterized by a neural network with a single hidden layer containing 512 units. Following the standard affine coupling mechanism, the 64-dimensional latent vector is partitioned into two equal 32-dimensional subsets. During each transformation, one subset remains unchanged and serves as the conditioning variable, while the other subset undergoes an affine transformation conditioned on the fixed subset. The transformed and fixed subsets are alternated across coupling layers to ensure full-dimensional expressiveness and invertibility. All implementation details are provided to facilitate reproducibility of the experimental results.

3. Additional Experiment: Extreme Missing-View Evaluation

Robustness under Extreme Missing-View Scenarios. To further investigate the behavior of DFL in more challenging settings, we evaluate a complete-view removal scenario, where one entire modality is unavailable during testing. The results are illustrated in Figure S1. It can be observed that the performance of all methods decreases when a full view is removed; however, DFL consistently maintains superior ACC, NMI, and ARI scores compared with BURG on both CiteSeer and Animal. More

importantly, the performance gap between DFL and BURG remains stable across different missing rates, indicating that DFL can effectively exploit the semantic relationships among views to recover missing information even when an entire modality is absent. These findings demonstrate the strong cross-view inference capability of the proposed flow-based latent alignment framework.

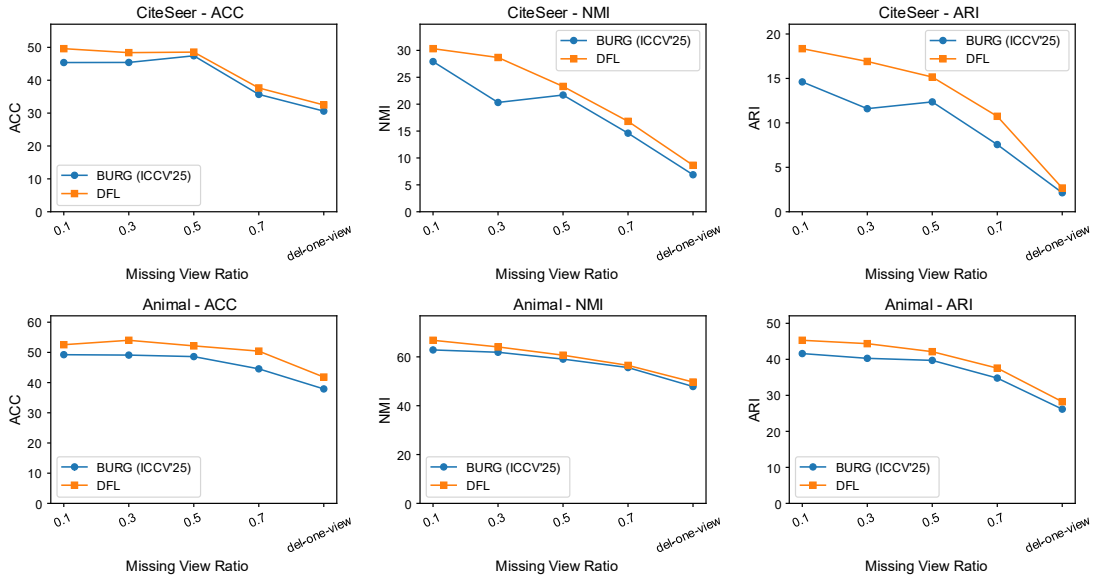


Figure S1: Performance comparison under complete-view removal on CiteSeer and Animal.

4. Additional Experiment: Robustness under Non-IID Distributions

Robustness to Non-IID Data Distributions. We further evaluate the robustness of DFL under non-IID conditions by injecting random noise into the observed views while preserving the original missing-view configurations. Figure Y presents the corresponding results on CiteSeer. As expected, both methods exhibit performance degradation under noisy distributions. Nevertheless, DFL consistently outperforms BURG across all missing rates and evaluation metrics. Furthermore, the overall performance trend of DFL remains relatively stable as the missing rate increases, suggesting that the learned latent representations are less sensitive to distribution shifts and observation noise. This robustness can be attributed to the multi-level structural alignment mechanism, which encourages semantically consistent latent representations and improves generalization in realistic non-IID environments.

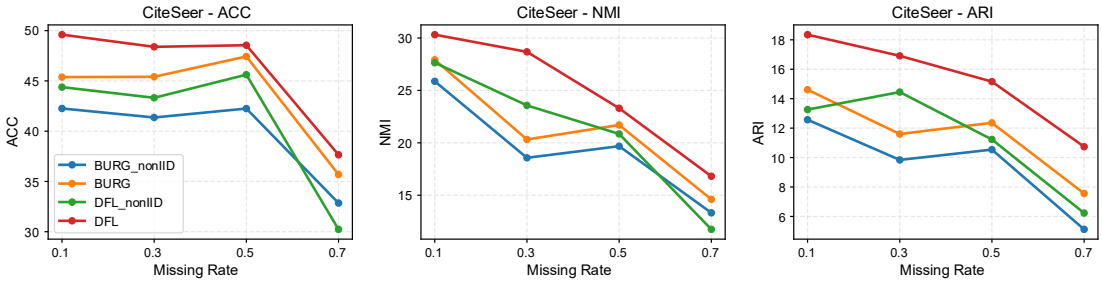


Figure S2: Performance comparison under Non-IID distributions on CiteSeer.

5. Computational Efficiency and Memory Consumption

We further evaluate the computational efficiency of DFL in terms of running time and resource consumption. The results are reported in Table S2 and Table S3. From Table

S2, we observe that DFL maintains competitive efficiency on most datasets compared with existing baselines. Although DFL incurs higher runtime on large-scale datasets such as Reuters and Youtube due to the additional conditional flow modeling and cross-view alignment operations, it remains within an acceptable computational range considering the significant performance gains. On smaller datasets, such as CUB and CiteSeer, DFL achieves comparable or even lower runtime than several strong baselines, demonstrating its efficiency in moderate-scale scenarios. Table S3 further reports the GPU memory consumption across different methods. Overall, DFL exhibits moderate memory usage compared to other deep multi-view clustering approaches. While the memory cost is slightly higher than lightweight models such as DIVIDE, it is comparable to methods with similar model capacity (e.g., BURG and ICMVC). This trade-off is expected, as DFL introduces conditional normalizing flows and multi-level alignment modules to improve representation quality. Overall, these results indicate that DFL achieves a reasonable balance between computational cost and clustering performance, making it suitable for a wide range of datasets and application scenarios.

6. Parameter Analysis

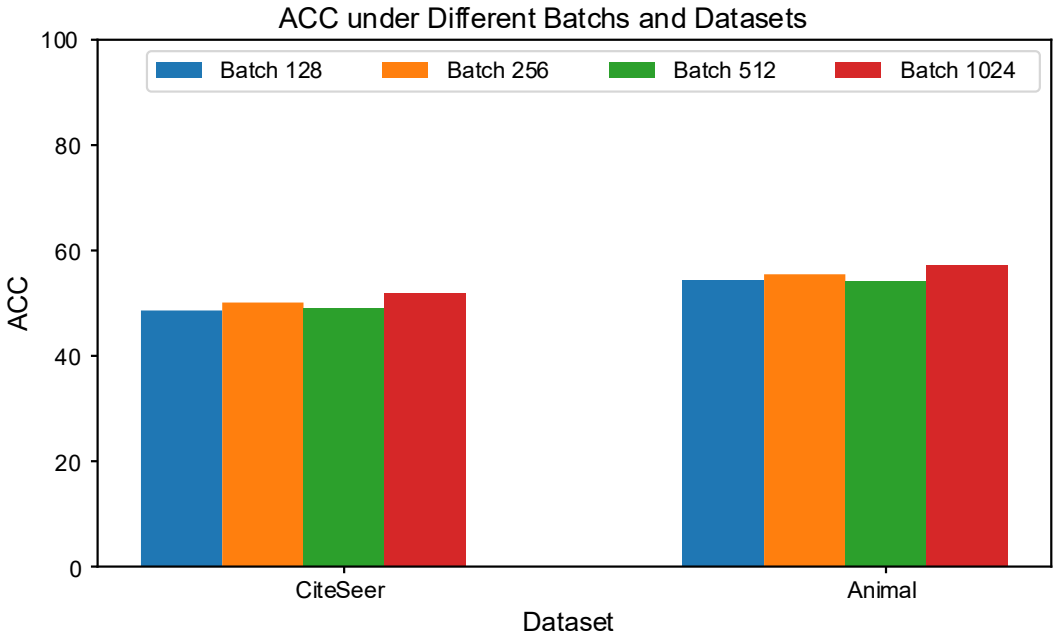


Figure S3: Sensitivity analysis of batch size on CiteSeer and Animal.

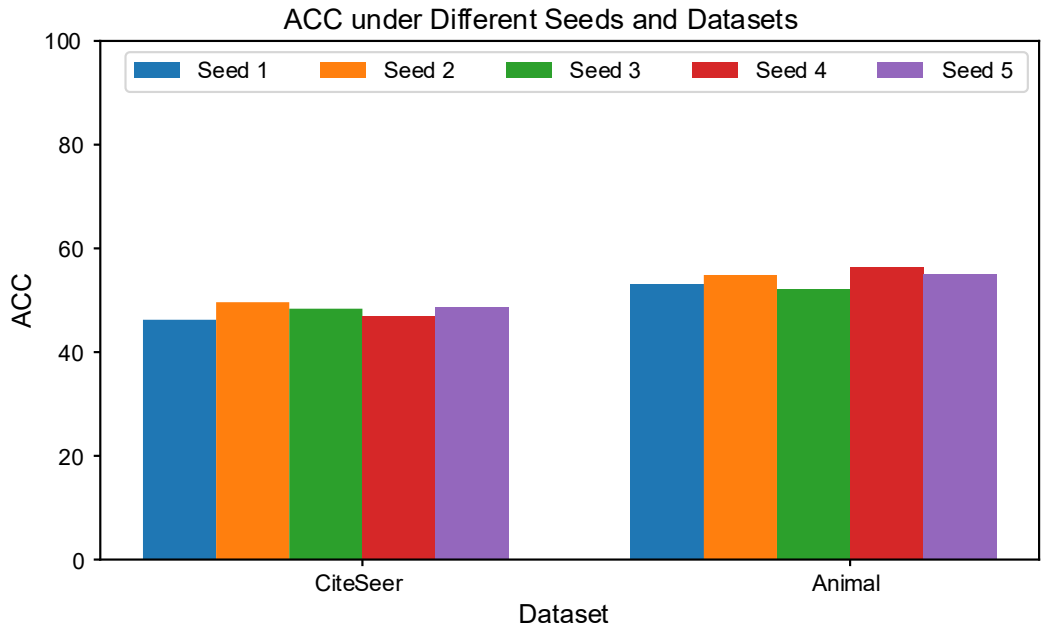


Figure S4: Sensitivity analysis of K-means initialization with different random seeds on CiteSeer and Animal.

Table s1. Clustering performance comparison across four datasets under four different missing rates. The best results are highlighted in bold, while the second-best results are underlined.

	Missing Rates	0.1			0.3			0.5			0.7		
	Metrics	ACC(%)	NMI(%)	ARI(%)	ACC(%)	NMI(%)	ARI(%)	ACC(%)	NMI(%)	ARI(%)	ACC(%)	NMI(%)	ARI(%)
Animal	COMPLETER (CVPR'21)	29.81	46.66	17.83	24.54	39.20	15.30	20.84	34.82	13.58	15.78	30.62	10.54
	DCP (TPAMI'22)	28.86	47.14	20.00	28.14	47.04	18.84	26.34	44.00	16.25	24.79	43.62	14.98
	SURE (TPAMI'22)	38.23	49.14	25.48	36.38	48.92	24.08	35.94	47.06	23.70	34.03	44.34	21.55
	DIMVC (AAAI'22)	48.29	60.20	35.79	47.74	59.72	37.71	46.42	56.31	33.46	45.27	53.06	28.31
	APADC (TIP'23)	30.23	48.44	25.01	29.15	46.72	23.31	28.79	45.51	22.72	27.33	43.39	20.79
	SMILE (TPAMI'23)	48.46	60.78	39.48	46.85	57.65	36.80	45.31	54.67	34.10	40.90	53.83	30.70
	DIVIDE (AAAI'24)	49.18	61.64	38.58	42.70	53.40	30.45	38.51	49.91	26.37	34.42	45.42	20.89
	ICMVC (AAAI'24)	45.10	60.73	33.62	41.17	57.19	31.55	38.34	54.30	28.76	34.67	50.71	25.14
	BURG (ICCV'25)	49.24	62.84	41.60	49.11	61.89	40.28	48.61	59.08	39.71	44.55	55.62	34.80
	DFL	**52.55**	**66.79**	**45.29**	**54.01**	**64.08**	**44.36**	**52.16**	**60.68**	**42.13**	**50.42**	**56.51**	**37.56**
Reuters	COMPLETER (CVPR'21)	39.54	21.43	12.81	37.59	20.85	11.77	32.76	13.03	8.46	26.97	8.32	4.81
	DCP (TPAMI'22)	43.51	23.95	16.96	41.89	22.95	15.97	40.19	21.64	14.42	39.03	20.91	13.23
	SURE (TPAMI'22)	41.32	25.66	13.32	40.44	22.40	10.59	37.63	21.28	7.58	37.03	18.21	6.63
	DIMVC (AAAI'22)	43.47	24.90	18.33	42.51	24.42	17.94	41.91	21.72	17.76	40.81	19.99	13.81
	APADC (TIP'23)	39.88	21.51	20.11	36.80	20.14	18.76	31.72	17.42	15.79	29.99	16.48	15.08
	SMILE (TPAMI'23)	42.39	23.78	15.43	39.03	22.51	15.02	36.08	20.51	14.04	35.58	18.98	12.03
	DIVIDE (AAAI'24)	40.90	23.64	18.96	39.59	22.55	14.78	38.01	21.62	15.81	37.86	20.98	15.01
	ICMVC (AAAI'24)	42.95	26.28	20.72	40.89	23.89	19.25	39.48	21.01	16.65	37.64	22.81	14.17
	BURG (ICCV'25)	44.01	26.39	21.15	42.86	26.38	20.22	42.97	24.77	19.61	41.84	23.63	20.60
	DFL	**49.21**	**28.67**	**25.61**	**48.41**	**27.87**	**24.22**	**49.88**	**26.48**	**24.93**	**44.47**	**24.42**	**21.22**
LandUse-21	COMPLETER (CVPR'21)	27.37	32.91	14.25	26.56	31.17	11.15	21.41	27.24	10.06	25.03	29.04	11.50
	DCP (TPAMI'22)	26.19	31.20	13.02	25.48	30.18	11.08	21.52	26.32	11.11	22.18	27.00	10.13
	SURE (TPAMI'22)	28.00	32.38	13.10	22.43	27.68	9.99	23.33	27.56	10.28	23.14	25.61	9.13
	DIMVC (AAAI'22)	22.99	29.78	10.56	22.05	27.60	9.76	24.20	28.51	10.21	23.63	27.40	9.36
	APADC (TIP'23)	20.92	26.74	7.97	19.99	24.40	7.44	19.15	23.91	7.08	18.97	21.44	6.78
	SMILE (TPAMI'23)	24.62	28.05	10.54	25.48	28.41	10.27	26.19	29.15	11.51	25.33	26.41	10.21
	DIVIDE (AAAI'24)	23.53	29.81	10.47	23.00	29.52	11.05	23.14	29.13	9.90	25.05	28.10	10.37
	ICMVC (AAAI'24)	26.81	30.71	13.60	26.12	29.83	12.87	24.43	27.27	11.12	22.97	25.15	9.65
	BURG (ICCV'25)	25.57	30.28	12.61	25.10	29.35	11.92	24.24	29.17	11.32	23.81	28.67	10.38
	DFL	**29.57**	**36.53**	**15.62**	**28.38**	**34.17**	**13.17**	**27.62**	**33.00**	**12.58**	**25.33**	**30.94**	**11.50**
YouTube	COMPLETER (CVPR'21)	21.30	10.04	4.91	18.10	7.48	2.14	21.90	10.64	4.20	20.51	9.04	3.54
	DCP (TPAMI'22)	20.88	11.02	3.99	20.45	10.48	4.08	21.19	10.15	4.02	20.42	9.25	2.90
	SURE (TPAMI'22)	22.55	12.11	5.59	24.55	11.78	6.30	22.40	9.67	5.30	22.00	8.72	4.23
	DIMVC (AAAI'22)	21.71	10.63	5.27	22.09	9.97	5.06	21.63	9.65	4.88	17.43	6.32	3.23
	APADC (TIP'23)	16.60	5.11	1.25	16.85	5.90	1.14	19.80	6.96	3.31	16.50	5.93	2.58
	SMILE (TPAMI'23)	23.50	9.94	5.22	22.20	9.82	5.21	23.50	9.02	4.97	21.60	8.68	4.28
	DIVIDE (AAAI'24)	26.90	13.40	6.34	24.25	11.40	6.09	27.20	13.18	6.50	25.15	10.23	5.39
	ICMVC (AAAI'24)	26.00	13.13	7.38	25.80	12.91	7.48	23.05	10.02	5.34	18.92	7.34	3.31
	BURG (ICCV'25)	26.05	16.81	8.49	26.60	15.79	9.13	22.10	12.59	5.76	21.85	11.29	5.46
	DFL	**30.75**	**20.45**	**12.17**	**29.50**	**18.37**	**10.86**	**30.65**	**18.57**	**9.56**	**29.00**	**16.91**	**8.18**

Table S2. Running time (seconds) comparison of different methods across multiple datasets.

Methods	CUB	Handwritten	CiteSeer	Animal	Reuters	LandUse-21	Youtube
COMPLETER	179.89	551.73	830.23	2595.00	4731.99	824.05	755.17
DCP	100.12	706.14	862.66	2496.44	10035.83	678.34	656.32
SURE	104.17	258.77	455.27	1357.81	1708.83	272.90	279.06
DIMVC	304.68	665.07	651.53	1190.73	1491.86	414.69	790.01
APADC	50.26	93.77	121.71	292.21	423.84	859.45	82.89
SMILE	138.15	349.35	518.37	1693.35	2135.91	320.11	351.82
DIVIDE	406.98	388.82	716.97	683.59	829.88	524.34	607.47
ICMVC	131.24	509.47	295.82	878.01	1445.18	256.62	225.94
BURG	46.06	495.52	416.54	1062.87	4074.71	281.62	531.04
DFL	144.00	2990.39	682.72	6438.35	18997.60	1668.36	2843.56

Table S3. GPU memory consumption (MB) comparison of different methods across multiple datasets.

Methods	CUB	Handwritten	CiteSeer	Animal	Reuters	LandUse-21	Youtube
COMPLETER	222.29	177.22	637.78	1372.13	272.05	175.67	344.82
DCP	217.00	270.99	635.77	1387.59	508.31	258.86	417.70
SURE	330.05	278.86	582.31	667.60	276.67	276.28	410.51
DIMVC	-	-	-	-	-	-	-
APADC	914.79	862.20	1328.20	1835.17	133.35	112.16	1031.85
SMILE	330.08	278.96	582.47	668.05	277.55	276.37	410.61
DIVIDE	160.69	207.24	368.37	395.96	204.43	206.28	276.52
ICMVC	185.64	402.02	621.72	3794.23	1948.27	361.90	471.25
BURG	227.51	645.76	259.16	447.78	486.79	240.19	772.90
DFL	207.87	824.05	415.01	458.88	859.45	351.87	1039.42